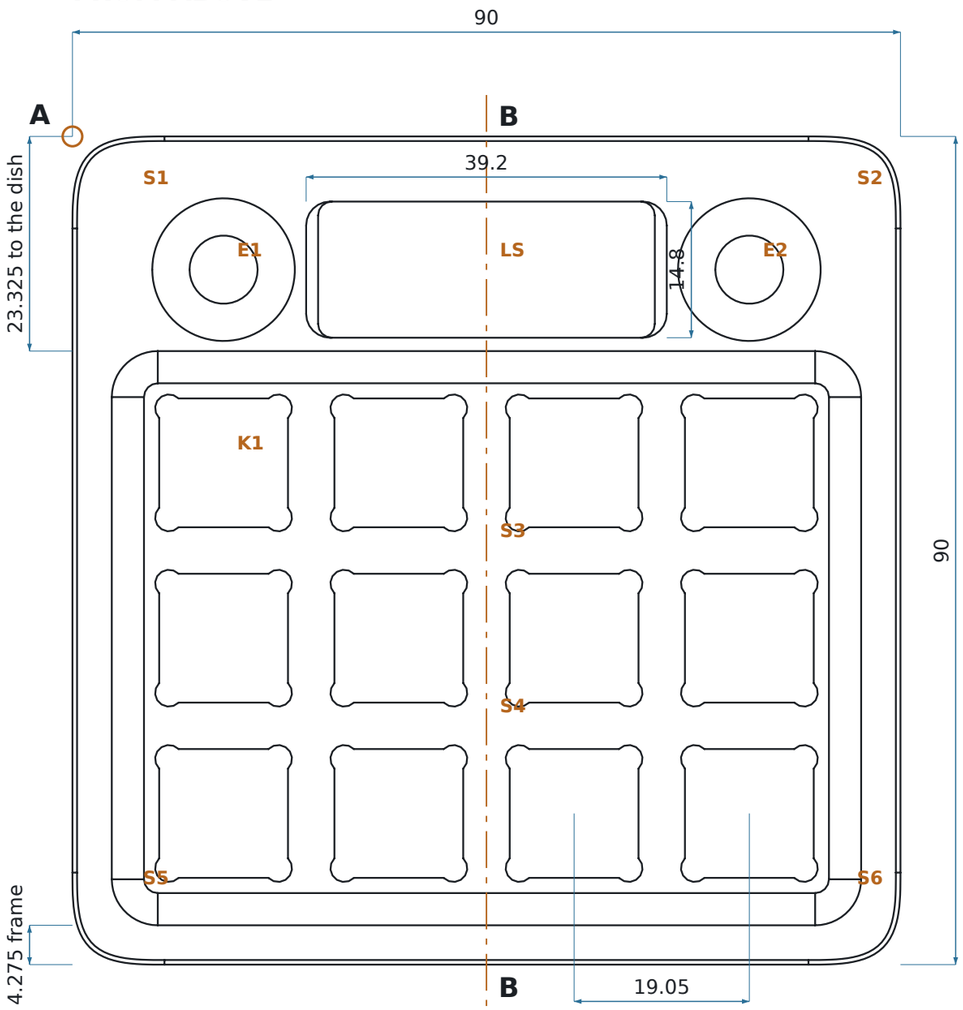
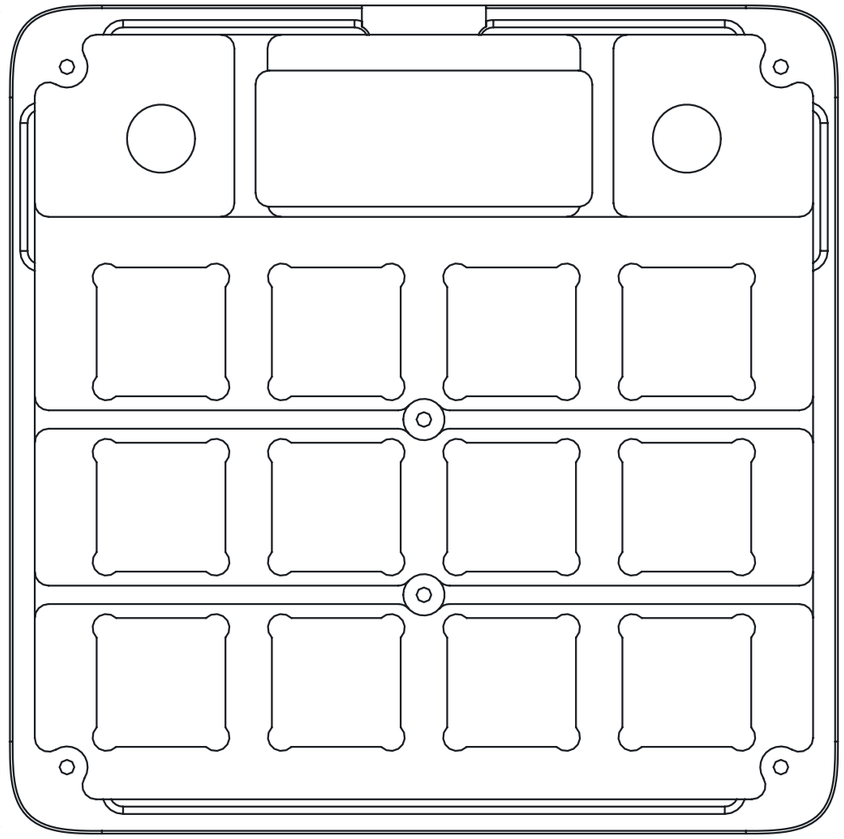
FROM ABOVE



FROM BELOW (mirrored left to right)

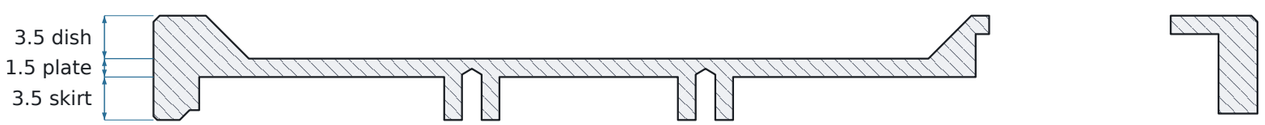


macropad case: top, CNC variant

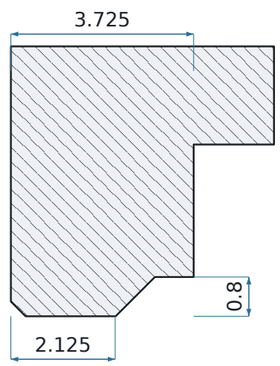
Holes and recesses, from datum A

tag	X	Y
ø1.6 x 3.7 from below (point to 4.18), tap M2 to 3 full thread		
S1	6.2	6.675
S2	83.8	6.675
S3	45	45
S4	45	64.05
S5	6.2	82.775
S6	83.8	82.775
ø7.4 through; well ø15.5 x 0.7 deep from above		
E1	16.425	14.475
E2	73.575	14.475
14 square through, first of 12 on a 19.05 grid		
K1	16.425	35.475
lens recess 39.2 x 14.8, R2.6, 1.25 deep from above		
opening 36.6 x 14.8 through, in its middle		
LS	45	14.475

SECTION B-B



GAP AND LAP JOINT, front wall, 8:1



Scale 1.5:1 plans, 2:1 section, 8:1 detail, on A3. Dimensions in mm, computed from the model.
Datum A: back-left corner of the box round the outline, seen from above. X to the right, Y toward the front.
3-axis, two setups: underside up, then top side up.
Every pocket corner is R1.5 or larger; the switch cut-outs have corner reliefs of that radius.
Switch cut-outs 14 square in a 1.5 mm plate. Take their tolerance from the Cherry MX datasheet.
Key dish: flat floor on the plate, walls at 45 degrees, 3.5 deep. A 90 degree chamfer mill of 8 mm or more cuts the walls.
Screw holes: drill 1.6 to 3.7 at full diameter, tap M2 with a bottoming tap to 3 full thread or more, for M2 x 8.
Setup 2 (top side up): hold the part on a fixture plate through the six M2 holes, or in soft jaws cut to the outline. The skirts are too thin for vise pressure.
Outline corners: one quintic Bezier each, control points 10, 6 and 2 mm from the box corner on both sides. Take the curve from the STEP file.
General tolerance: ISO 2768-m is a proposal. Nothing has been made or measured.